

Exact SmallGroups Frontiers for Three Questions on Central Fourier Amenability Constants

Run `fa_banach_001` — lane 16

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Status: candidate computational partial result, likely valid. An exact exhaustive GAP computation proves affirmative answers to Questions 6.2–6.4 of arXiv:2210.16262 for every finite group of order at most 383. This is a finite theorem, not a structural solution of any of the three unrestricted questions. It depends on the correctness of GAP’s character-table and SmallGroups data and should receive independent human reproduction.

1 Source questions

John Sawatzky, *Amenability constants of central Fourier algebras of finite groups*, arXiv:2210.16262v1 (2022), asks on PDF page 16 [1]:

- 6.2. Does $\text{AM}_{ZA}(G/N) \leq \text{AM}_{ZA}(G)$ when N is a normal Hall subgroup?
- 6.3. If $\text{AM}_{ZA}(G) = 7/4$, is it necessary that G be nilpotent?
- 6.4. Is $\text{AM}_{ZA}(G') \leq \text{AM}_{ZA}(G)$, where G' is the derived subgroup?

The source notes that all groups of order below 192 respect every quotient, then exhibits a non-Hall quotient counterexample at order 192.

monomial groups of order less than 100, and 21 of them have the property that $\text{AMZA}(G) \neq \text{AMZL}(G)$.

Question 6.2. Does $\text{AMZA}(G/N) \leq \text{AMZA}(G)$ hold if N is a normal Hall subgroup of G ?

Question 6.3. For which groups is $\text{AMZA}(G) = \frac{7}{4}$ actually achieved? Is it necessary that G be nilpotent?

Question 6.4. Is it true that $\text{AMZA}(G') \leq \text{AMZA}(G)$, where G' is the derived subgroup of G ?

Question 6.5. Are there other gaps respected by amenability constants besides around 1?

2 Result

Theorem 1 (Exact finite frontier). *For every finite group G with $|G| \leq 383$:*

1. if $N \triangleleft G$ is a Hall subgroup, then $\text{AM}_{ZA}(G/N) \leq \text{AM}_{ZA}(G)$;
2. if $\text{AM}_{ZA}(G) = 7/4$, then G is nilpotent;
3. $\text{AM}_{ZA}(G') \leq \text{AM}_{ZA}(G)$.

All comparisons in this theorem were performed with exact algebraic values, not floating-point approximations.

3 Proof intuition

The source expresses $\text{AM}_{ZA}(G)$ directly in terms of the irreducible character table and conjugacy-class sizes. This makes the three bounded problems finite and exact: enumerate every `SmallGroups` identifier, construct only the subgroup or quotient in question, and compare the resulting cyclotomic expressions.

Two safe filters greatly reduce the expensive cases. An abelian subgroup or quotient has amenability constant 1, so it cannot violate either monotonicity inequality. For the $7/4$ question, the diagonal part of the character sum is the exact lower bound $\text{ass}(G) \leq \text{AM}_{ZA}(G)$; only groups with $\text{ass}(G) \leq 7/4$ need the full calculation.

4 Exact formula and exhaustive algorithm

Write $\text{Irr}(G)$ for the irreducible characters, $d_\chi = \chi(1)$, and $\text{Conj}(G)$ for the conjugacy classes. The formula used is

$$\text{AM}_{ZA}(G) = \frac{1}{|G|^2} \sum_{\chi, \psi \in \text{Irr}(G)} d_\chi d_\psi \left| \sum_{C \in \text{Conj}(G)} |C|^2 \chi(C) \overline{\psi(C)} \right|. \quad (1)$$

The diagonal minorant is

$$\text{ass}(G) = \frac{1}{|G|^2} \sum_{\chi} d_\chi^2 \sum_C |C|^2 |\chi(C)|^2 \leq \text{AM}_{ZA}(G). \quad (2)$$

The supplied GAP program implements (1) with exact character values. Each absolute value is evaluated as `Sqrt(s*ComplexConjugate(s))`; no decimal conversion occurs before a comparison. It then performs these exhaustive scans.

1. For Question 6.2, every proper nontrivial prime subset is considered. For solvable G , Hall's conjugacy theorem [2] shows that one Hall subgroup for that subset suffices: if a normal Hall subgroup exists, every conjugate is the same subgroup. For nonsolvable G , the code enumerates all normal subgroups and tests the Hall condition. Abelian quotients are omitted.
2. For Question 6.3, every nonnilpotent G is tested first against (2); the exact full value is computed for every survivor.
3. For Question 6.4, G' is constructed for every G ; only cases with nonabelian G' require a full comparison.

The `SmallGroups` library is complete for every order in the tested interval. Thus every finite group of order at most 383 occurs, up to isomorphism, and the preceding loops prove Theorem 1, subject only to the stated software/data trust boundary.

5 Counts and validation

The computation used GAP 4.15.1 and SmallGrp 1.5.4 [3, 4].

scan	cases before filter	exact full comparisons	violations
normal Hall quotients	7,747 relevant quotients	7,747	0
$\text{AM}_{ZA}(G) = 7/4$, nonnilpotent G	6,892 groups	186	0
derived subgroup	all groups $ G \leq 383$	976	0

Here “relevant” means that the quotient or derived subgroup is nonabelian. The largest single family was order 256: all 56,092 groups were enumerated, and 410 required the derived-subgroup comparison.

As an implementation gate, the program reproduced six known/source values:

(G , id)	$(6, 1)$	$(8, 3)$	$(24, 12)$	$(32, 43)$	$(96, 204)$	$(192, 1022)$
$\text{AM}_{ZA}(G)$	$7/3$	$7/4$	$511/72$	$83/32$	$497/32$	$1727/128$

In particular, the last two are exactly 15.53125 and 13.4921875, matching the quotient counterexample in the source.

The full program is `code/2210_16262_amza_scan.g`; the chunked commands and terminal summaries are in `code/verification_output.txt`.

6 Limitations and upgrade attempts

This packet does not claim any unrestricted question is solved. Four focused attempts were made: exact formula validation, the Hall scan, the derived scan, and a separate $7/4$ scan using the source minorant. Clifford-theoretic reorganization for a normal Hall semidirect product did not yield termwise domination in (1); restriction to G' similarly introduces both character multiplicities and conjugacy-class splitting. Equality in the off-diagonal sum also lacks a structural characterization. These are the current barriers to promoting the finite result to a proof for all groups.

Bounded searches of the run indexes, the source arXiv id and title, and exact phrases from Questions 6.2–6.4 found the source paper but no later claimed answer. This is not a comprehensive literature review. Novelty confidence is moderate, and the recommended next step is independent reproduction plus expert review of the completeness/filter arguments.

References

- [1] J. Sawatzky, *Amenability constants of central Fourier algebras of finite groups*, arXiv:2210.16262v1, 2022.
- [2] P. Hall, *A note on soluble groups*, J. London Math. Soc. 3 (1928), 98–105.
- [3] The GAP Group, *GAP – Groups, Algorithms, and Programming*, Version 4.15.1.
- [4] H. U. Besche, B. Eick, and E. A. O’Brien, *The SmallGroups Library*, GAP package SmallGrp, Version 1.5.4.